Project Design Phase-II
Data Flow Diagram & User Stories

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Project Name	Strategic product placement analysis: unveiling sales impact with tableau visualization

Data Flow Diagrams:

Data Flow Architecture

1. Data Collection & Extraction (The Inbound Layer)

Sources: Raw datasets are gathered from Kaggle, focusing on variables like Product ID, Position, and Sales Volume.
Connection: The systematically gathered data is imported into Tableau Desktop, linking relevant columns to established data fields.

2. Data Preparation (The Processing Layer)

Cleaning & Transformation: Inaccuracies are removed, and raw variables are structured to ensure software compatibility.
Filtering: Analysts refine the scope by focusing on subsets like Demographics or Seasonality to address specific objectives.

3.Data visualization& Dashboard (The Consumption Layer)

Visual Logic: Processed data flows into unique visualizations (e.g., Avg Sales Volume vs. Product Category) to uncover patterns.
Consolidation: Individual sheets are pulled into a Responsive Dashboard using a floating layout for clear, interactive analysis.

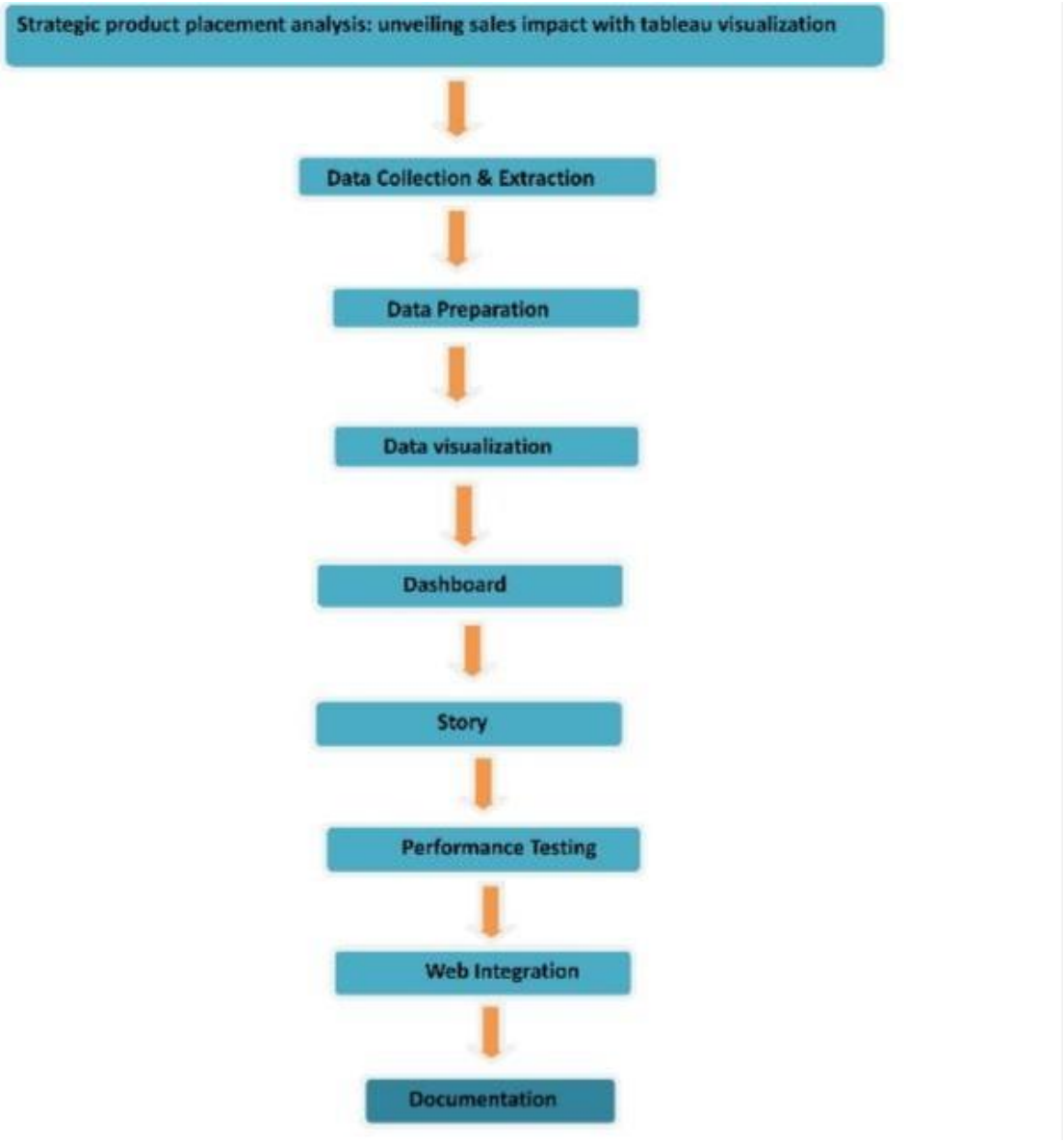
4. Story & Performance Testing (The Insight Layer)

Narrative Flow: The dashboard results move into a Story, breaking the analysis into a sequence of scenes to explain "Sales Impact".
Optimization: Performance testing ensures that Filters and Calculation Fields (like Revenue and ROI) operate efficiently under load.

5. Web Integration & Documentation (The Delivery Layer)

Publishing: The final workbook is saved to Tableau Public, where an embedded code is generated
Deployment: The embedded code is integrated into a Bootstrap template and served via Flask, making the project accessible as a web solution.

Closure: The entire end-to-end flow is recorded in an explanation video and step-by-step documentation.



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Retail	Spatial	USN-1	As a manager, I	I can	High	Sprint

Manager	Dashboard		want to see a Heat Map of the store floor.	identify "Hot" and "Dead" zones visually.		-1
Retail Manager	Profit Analysis	USN-2	As a manager, I want to compare Eye-level vs. Bottom-shelf profit.	I can see a bar chart showing the margin difference.	High	Sprint -1
Category Analyst	Data Integration	USN-3	As an analyst, I want to upload POS CSV files into the system.	Data is cleaned and mapped to store coordinates.	High	Sprint -1
Category Analyst	Filtering	USN-4	As an analyst, I want to filter sales by Product Category	I can see how "Snacks" perform vs "Dairy" in specific aisles.	Med	Sprint -2
Administrator	Layout Config	USN-5	As an admin, I want to update the store floor plan (X/Y map).	The Tableau map updates to reflect new aisle positions.	Med	Sprint -2
Executive	Reporting	USN-6	As an executive, I want a Weekly Sales Lift report.	I receive an automated email with total revenue impact.	Low	Sprint -2